

CSC1006 - Final Reflection Paper

Due time: 2026/05/10 23:59:59

Choose one or more topics from the list below, read or search for the relevant materials, and write an essay of **longer than 1000 words**:

- Nobel Prize in Physics, 2024. The Nobel Prize in Physics 2024 was awarded jointly to John J. Hopfield and Geoffrey Hinton “for foundational discoveries and inventions that enable machine learning with artificial neural networks”. Read the relevant information from the official Nobel Prize website [1] and write down your reflection.
- AI for scientific discovery. Read an article on how AI can accelerate scientific discovery [2] and write down your reflection.
- AI for math. Google DeepMind AI published AlphaGeometry 2, which reaches the level of gold-medal students in the International Mathematical Olympiad. Read relevant news and posts on the Internet and write down your reflection.
- AI for research. Read an article on how AI is used in the research workflow [3], as well as other relevant materials on the Internet, and write down your reflection.
- One detailed AI application example in commercialization (e.g. industry, medicine, transportation), could be a software system, or a hardware device like robots, etc., to explain its development history, detailed method, and current state-of-the-art performance.
- Any in-depth thinking of an AI-relevant object in reality (could be a game, a movie, or a social event), should introduce the object itself, and to add your well-structured and logical analysis about it.

[1] <https://www.nobelprize.org/prizes/physics/2024/summary/>

[2] <https://www.nature.com/articles/s42254-022-00518-3>

[3] <https://www.nature.com/articles/d41586-025-00343-5>

Notices:

- **Any reflection content that is positive and constructive to the reality is encouraged. But be sure to avoid extreme and/or inappropriate insights.**
- **MUST NOT directly use AI-generated content, those caught will get 0 scores.**
- **Please avoid late submission. Late submission within 24 hrs will get 50% score deducted, and no scores for later than 48 hrs.**